Software Requirements Specification (SRS)

For

Optical Service Web System

HNDIT – 4052

Programming Individual Project



Version 1.0

Prepared by:

RAT/IT/2324/F/0018

W.P.M.Malindi Pabasara

Under the Supervision of :

Mr. G.S.T. Bandara

Date : 04.05.2026

This document has been prepared as a part of an academic project. Any opinions, finding and conclusions expressed in this document are those of the author and do not necessarily reflect the views of university or organization.

Acknowledgement

I would like to express my sincere gratitude to my supervisor, Mr. Thilina Bandara, for his continuous guidance, support, and valuable advice throughout the development of this project. His expertise and encouragement greatly helped me in completing this Software Requirements Specification successfully.

I would also like to thank the lecturers of the Department of Information Technology at the Advanced Technological Institute, Rathnapura, for providing me with the knowledge and skills necessary to carry out this project.

Furthermore, I extend my appreciation to my friends and colleagues for their support, ideas, and encouragement during the development process.

Finally, I would like to thank my family for their understanding, motivation, and continuous support, which helped me to successfully complete this project.

Table of Contents

Page

Acknowledgement…….. 02

Table of Contents…… 03

Revision History…….. 04

1. **Introduction……. 05**
   - 1.1 Purpose…… 05
   - 1.2 Document Conventions……..… 05
   - 1.3 Intended Audience and Reading Suggestions…….. 06
   - 1.4 Project Scope…… 06
   - 1.5 References…… 07
2. **Overall Description…….. 08**
   - 2.1 Product Perspective…….. 08
   - 2.2 Product Features……... 08
   - 2.3 User Classes and Characteristics…….. 09
   - 2.4 Operating Environment……..…. 10
   - 2.5 Design and Implementation Constraints……... 11
   - 2.6 User Documentation……..….. 12
   - 2.7 Assumptions and Dependencies……..... 13
3. **System Features……..…... . 14**
   - 3.1 User Registration and Login ……..…… 14
   - 3.2 Appointment Booking System…….……..…….. 14
   - 3.3 Prescription Management ……...……..……... 14
   - 3.4 Product Catalog(Glasses & Lenses) ……..……...15
   - 3.5 Online Shopping and Checkout ……..……..15
   - 3.6 Order Tracking System ……..…….. 15
   - 3.7 Admin Management…… ……..…….. 16
4. **External Interface Requirements……..…….. 17**
   - 4.1 User Interfaces……..…… 17
   - 4.2 Hardware Interfaces…….…… 17
   - 4.3 Software Interfaces……..……. 18

4.4 Communications Interfaces…… 18
5. Other Nonfunctional Requirements…….. 19
5.1 Performance Requirements……. 19
5.2 Safety Requirements……... 19
5.3 Security Requirements…… 19
5.4 Software Quality Attributes…… 20
Appendix A: Glossary……... 21

Revision History

Name	Date	Reason For Changes	Version
Malindi Pabasara	2026.05.04	Initial draft of SRS document based on IEEE Std 830-1998	1.0

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed description of the requirements for the Optical Service Web System. This document outlines the system's functional and non-functional requirements, system features, and overall behavior to ensure a clear understanding between stakeholders, developers, and users.

It serves as a guideline for the design, development, testing, and implementation of the system, ensuring that all requirements are clearly defined and properly addressed. Additionally, this document acts as a reference for future system maintenance and enhancements.

1.2 Document Conventions

The following conventions are used in this Software Requirements Specification (SRS) document to ensure clarity, consistency, and ease of understanding.

Heading Style: All major and minor headings are structured in a numbered format (e.g., 1, 1.1, 1.2) and are written in bold for clear identification and organization of content.

Requirement Format: All system requirements are written using the standard format "The system shall…" to maintain clarity, precision, and consistency throughout the document.

Requirement Identification: Each requirement is assigned a unique identifier (e.g., R-LOGIN-01, R-APPT-02) to facilitate easy tracking, referencing, and management.

Text Style: All descriptions are written in clear, simple, and formal academic language to ensure the document is understandable to both technical and non-technical readers.

Bullet Points: Bullet points and numbered lists are used to present features and requirements in a structured and readable manner.

Consistency: Uniform terminology and naming conventions are maintained throughout the document to avoid ambiguity and misinterpretation.

These conventions ensure that the document is well-structured, easy to read, and suitable for academic and professional evaluation.

1.3 Intended Audience

This Software Requirements Specification (SRS) document is intended for developers, system designers, testers, project managers, and stakeholders involved in the development of the Optical Service Web System. It is also useful for system administrators and future maintainers who will work with or support the system after deployment.

Readers are encouraged to go through the document in sequence, starting from the Introduction to gain a general understanding of the system, followed by the Overall Description and System Features for detailed requirements. Technical users such as developers and testers should focus on the functional and non-functional requirements, while non-technical stakeholders can refer to the overview sections for a general understanding of the system.

1.4 Project Scope

The Optical Service Web System is a web-based application designed to modernize and improve the management of optical services. The system will provide an integrated platform that allows customers to book eye test appointments online, access their prescription records, and browse and purchase optical products such as glasses and contact lenses.

The system will also support administrative functions, including managing customer information, appointments, prescriptions, products, and orders. It aims to reduce manual work, minimize errors, and improve service efficiency and customer satisfaction.

This project focuses on developing a user-friendly, secure, and responsive web application that can be accessed through desktop and mobile devices. However, the scope is limited to core functionalities and does not include advanced features such as AI-based recommendations or integration with external medical systems.

1.5 References

The following documents and resources were used as references in the preparation of this Software Requirements Specification (SRS):

- Wikipedia OSMS page: https://en.wikipedia.org/wiki/Optician
- MySQL Official Document: MySQL
- React Official Website: https://react.dev

2. Overall Description

2.1 Product Perspective

The Optical Service Web System is a standalone web-based application designed to replace traditional manual and semi-digital optical service management processes. It provides an integrated platform that connects customers, optical service providers, and administrators within a single system.

The system will operate as a centralized solution where all data related to customers, appointments, prescriptions, products, and orders are stored and managed in a database. It will interact with standard web technologies such as web browsers and internet servers, making it accessible from various devices including desktops, laptops, and mobile phones.

This system is not a part of any existing software but can be expanded in the future to integrate with other systems such as payment gateways, inventory management systems, or healthcare platforms. Overall, it aims to modernize optical service operations by improving efficiency, accessibility, and data management.

2.2 Product Features

The Optical Service Web System provides the following key features:

User Registration and Login:

Allows customers and staff to create accounts and securely access the system.

Appointment Booking System:

Enables customers to schedule, reschedule, and manage eye test appointments online.

Prescription Management:

Stores and manages customer prescriptions digitally for easy access and updates.

Product Catalog:

Displays available optical products such as glasses and contact lenses with details.

Shopping Cart and Checkout:

Allows customers to select products, add them to a cart, and complete purchases.

Order Management:

Tracks customer orders, including order status and history.

Admin Dashboard:

Provides tools for administrators to manage users, products, appointments, and reports.

2.3 User Classes

The Optical Service Web System will have several types of users, each with different roles, responsibilities, and access levels within the system.

1. Customers

Customers are the primary users of the system. They can register, log in, book eye test appointments, view their prescriptions, and purchase optical products such as glasses and contact lenses. Customers are expected to have basic knowledge of using web applications and internet browsing.

2. Admin (System Administrator)

The admin has full control over the system. This includes managing users, products, appointments, prescriptions, and orders. The admin is responsible for maintaining the system, monitoring activities, and ensuring smooth operation. This user requires a higher level of technical knowledge and system understanding.

3. Optical Staff (Optician/Employee)

Optical staff are responsible for handling customer-related services such as managing appointments, updating prescriptions, and assisting in order processing. They have limited access compared to the admin and are expected to have moderate technical skills to operate the system efficiently.

Each user class has different access permissions to ensure system security and proper functionality. The system will provide a user-friendly interface suitable for all types of users, regardless of their technical background.

2.4 Operating Environment

The Optical Service Web System will operate in a modern web-based environment and will be accessible through multiple devices and platforms. The system is designed to ensure compatibility, flexibility, and smooth performance across different user environments.

1. Hardware Requirements

Desktop computers, laptops, tablets, and smartphones

Minimum 4GB RAM (recommended 8GB or higher for better performance)

Stable internet connection (broadband or mobile data)

2. Software Requirements

Operating Systems:

Windows 10/11

macOS

Android (for mobile access)

iOS (for mobile access)

Web Browsers:

Google Chrome (recommended)

Mozilla Firefox

Microsoft Edge

Safari

3. Server Environment

Web Server: Apache / Nginx

Backend: Node.js / PHP / Java (depending on your implementation)

Database Server: MySQL / MongoDB

4. Network Requirements

Internet connectivity is required for all users

Minimum bandwidth: 1 Mbps (recommended 5 Mbps or higher for smooth usage)

5. Development Environment (Optional for project documentation)

VS Code / IntelliJ IDEA / Eclipse

Git version control system

XAMPP / WAMP (for local testing)

2.5 Design and Implementation Constraints

The Optical Service Web System must be developed within certain technical, operational, and organizational limitations. These constraints influence how the system is designed and implemented.

1. Technology Constraints

The system must be developed using web technologies such as HTML, CSS, JavaScript, and a suitable backend language (e.g., PHP, Node.js, or Java).

The database must be implemented using a relational database management system such as MySQL.

The system must be compatible with modern web browsers (Chrome, Firefox, Edge, Safari).

2. Platform Constraints

The system must be web-based and accessible through desktop and mobile devices.

It must support cross-platform usage (Windows, macOS, Android, iOS) through browsers.

3. Performance Constraints

The system should load pages within a reasonable time (ideally under 3–5 seconds under normal network conditions).

It must handle multiple concurrent users without significant performance degradation.

4. Security Constraints

User authentication must be implemented for secure access (e.g., login system for customers and administrators).

Sensitive data such as passwords must be encrypted.

Secure communication (HTTPS) should be used.

5. Hardware Constraints

The system must run on standard server hardware with sufficient RAM and storage capacity.

Client devices only require a web browser and internet connection.

6. Development Time Constraints

The system must be completed within the academic semester timeline.

Limited time may restrict implementation of advanced features such as AI-based recommendations or advanced analytics.

7. Resource Constraints

Limited budget may restrict the use of paid tools, APIs, or cloud services.

Development will rely mainly on free or open-source technologies.

2.6 User Documentation

The Optical Service Web System will provide user documentation to help users understand how to effectively use the system. This documentation will include clear instructions on how to register, log in, book appointments, place orders, and track services. It will also guide administrators on how to manage customers, update inventory, handle orders, and generate reports. The user documentation will be written in simple and easy-to-understand language, supported with screenshots or diagrams where necessary. It will be available as an online help section within the system as well as a downloadable PDF user manual, ensuring that both technical and non-technical users can easily operate the system without confusion.

2.7 Assumptions and Dependencies

The Optical Service Web System is developed based on several assumptions and external dependencies to ensure proper functionality and successful deployment. It is assumed that users will have basic knowledge of using web browsers and access to a stable internet connection to interact with the system. The system also assumes that all users (customers and administrators) will provide accurate and valid information during registration and service usage. It is further assumed that the hosting environment will remain stable and that the system will be maintained regularly to avoid technical failures.

In terms of dependencies, the system relies on external technologies such as web browsers (Google Chrome, Firefox, Edge, Safari) and a stable internet service for communication between clients and the server. It also depends on backend technologies (such as PHP, Node.js, or Java) and a database management system like MySQL for data storage and processing. Additionally, the system may depend on third-party services such as email services for notifications and hosting platforms for deployment. Any failure or limitation in these dependencies may affect the overall performance and availability of the system.

3. System Features

3.1 User Registration and Login

Description:

This feature allows customers and admins to securely access the system.

Functional Requirements:

- ✓ R-LOGIN-01: The system shall allow users to register with personal details (name, email, password, contact number).
- ✓ R-LOGIN-02: The system shall allow users to log in using valid credentials.
- ✓ R-LOGIN-03: The system shall validate user credentials before granting access.
- ✓ R-LOGIN-04: The system shall allow users to log out securely.

3.2 Appointment Booking System

Description:

This feature allows customers to book eye test appointments online.

Functional Requirements:

- ✓ R-APPOINT-01: The system shall allow users to view available dates and time slots.
- ✓ R-APPOINT-02: The system shall allow users to book an appointment.
- ✓ R-APPOINT-03: The system shall store appointment details in the database.
- ✓ R-APPOINT-04: The system shall allow users to cancel or reschedule appointments.

3.3 Prescription Management

Description:

This feature allows storing and accessing customer eye prescriptions.

Functional Requirements:

- ✓ R-PRES-01: The system shall allow admin/optician to add prescriptions.
- ✓ R-PRES-02: The system shall store prescription history for each user.
- ✓ R-PRES-03: The system shall allow users to view their prescriptions.
- ✓ R-PRES-04: The system shall ensure secure access to prescription data.

3.4 Product Catalog (Glasses & Lenses)

Description:

This feature displays available optical products.

Functional Requirements:

- ✓ R-PROD-01: The system shall display a list of available products.
- ✓ R-PROD-02: The system shall show product details (price, type, description).
- ✓ R-PROD-03: The system shall allow users to search/filter products.

3.5 Online Shopping and Checkout

Description:

This feature allows users to purchase optical products.

Functional Requirements:

- ✓ R-CART-01: The system shall allow users to add products to the cart.
- ✓ R-CART-02: The system shall allow users to update/remove items from the cart.
- ✓ R-CART-03: The system shall allow users to checkout and place orders.
- ✓ R-CART-04: The system shall support secure payment methods.

3.6 Order Tracking System

Description:

This feature allows users to track their orders.

Functional Requirements:

- ✓ R-ORDER-01: The system shall generate an order after checkout.
- ✓ R-ORDER-02: The system shall allow users to view order status.
- ✓ R-ORDER-03: The system shall update order status (processing, shipped, delivered).

3.7 Admin Management

Description:

This feature allows admin to manage the system.

Functional Requirements:

- ✓ R-ADMIN-01: The system shall allow admin to manage users.
- ✓ R-ADMIN-02: The system shall allow admin to manage products.
- ✓ R-ADMIN-03: The system shall allow admin to manage appointments.
- ✓ R-ADMIN-04: The system shall allow admin to view reports and system activities.

4. External Interface Requirements

4.1 User Interface (UI)

The system shall provide a simple, responsive, and user-friendly web-based interface that supports both desktop and mobile users. The interface is designed to ensure easy navigation and efficient access to all optical services.

Key UI components include:

- Home Page: Displays services, products, promotions, and navigation menu.
- User Authentication Interface: Sign up and sign in pages for customers and administrators with secure login validation.
- Customer Dashboard: Shows profile, appointment history, orders, and notifications.
- Admin Dashboard: Allows management of users, appointments, inventory, and reports.
- Product Catalog Interface: Displays optical products (glasses, lenses, accessories) with images, prices, and descriptions.
- Appointment Booking Interface: Enables users to book eye tests and consultations by selecting date and time.
- Order Management Interface: Allows customers to place orders and track order status.
- Search and Filter Interface: Helps users easily find products and services.
- Notification Interface: Shows alerts for appointments, orders, and system updates.
- Contact/Support Interface: Allows users to submit inquiries and get assistance.

The UI shall ensure consistency, fast response time, and accessibility for all users.

4.2 Hardware Interface

The system will operate on standard hardware devices and does not require specialized hardware.

Supported hardware includes:

- ✓ Desktop computers and laptops for full system access
- ✓ Smartphones and tablets for mobile access (responsive design)
- ✓ Server hardware for hosting the web application and database
- ✓ Input devices such as keyboard and mouse

- ✓ Internet routers/modems for network connectivity

Minimum hardware requirements for server:

- Processor: Intel i5 or higher (or equivalent)
- RAM: 8 GB minimum
- Storage: 100 GB or more
- Network: Stable broadband internet connection

4.3 Software Interface

- The system will interact with the following software components:
- Operating Systems: Windows, Linux, macOS (server-side and client-side support)
- Web Browsers: Google Chrome, Mozilla Firefox, Microsoft Edge, Safari
- Web Server: Apache or Nginx
- Backend Technology: (e.g., PHP / Java / Node.js depending on your implementation)
- Database Management System: MySQL or MongoDB
- Frontend Technologies: HTML, CSS, JavaScript (possibly Bootstrap for UI design)
- Development Tools: VS Code, NetBeans, or similar IDE
- Version Control System: Git / GitHub (optional for collaboration)

4.4 Communication Interface

The system shall use standard internet-based communication protocols to ensure secure and reliable data transfer.

- HTTP/HTTPS Protocol: Used for communication between client and server
- REST API: Used for data exchange between frontend and backend systems
- JSON Format: Used for data transmission between services
- Email Service Integration: Used for sending notifications such as appointment confirmations and order updates
- Internet Connection: Required for accessing all system features
- Secure Socket Layer (SSL/TLS): Ensures secure communication and data encryption

5. Non-Functional Requirements

5.1 Performance Requirements

The system shall be designed to deliver fast and efficient performance to ensure a smooth user experience. All pages, including product listings, appointment booking, and dashboards, should load within a few seconds under normal network conditions. The system must be capable of handling multiple users simultaneously without significant performance degradation. Database queries and transactions should be optimized to ensure quick response times, especially during peak usage periods such as promotions or appointment bookings.

5.2 Safety Requirements

The system shall ensure that all user data and system operations are handled safely to prevent data loss or corruption. Regular database backups should be maintained to recover information in case of system failure or accidental deletion. The system should be designed to avoid critical errors that may affect users during transactions such as booking appointments or placing orders. In case of system failure, recovery mechanisms should restore the system to its last stable state with minimal disruption.

5.3 Security Requirements

The system shall implement strong security measures to protect user data and prevent unauthorized access. All user authentication processes must use secure login mechanisms with encrypted passwords. Sensitive data such as personal information and payment details should be encrypted during transmission using HTTPS/SSL protocols. The system should include role-based access control to ensure that only authorized users (e.g., admin, staff, customers) can access specific functionalities. Additionally, the system must protect against common security threats such as SQL injection, cross-site scripting (XSS), and unauthorized access attempts.

5.4 Software Quality Attributes

The Optical Service Web System shall ensure high-quality software characteristics to provide a reliable and user-friendly experience. The system shall maintain usability by offering an intuitive and easy-to-navigate interface suitable for all types of users, including customers and administrators. It shall ensure reliability by operating consistently without frequent failures and by properly handling errors during system usage. The system shall support maintainability, allowing developers to easily update, fix bugs, and add new features without affecting existing functionality. Scalability shall be considered so that the system can handle an increasing number of users, products, and transactions as the business grows. Additionally, the system shall ensure efficiency by optimizing resource usage and providing fast response times, and portability by supporting different web browsers and operating systems without compatibility issues.

7. Appendix A: Glossary

User

A customer who uses the system to book appointments, view prescriptions, and purchase products.

Admin

The system administrator who manages users, products, appointments, and system operations.

Optician

A staff member who manages eye tests and updates prescriptions.

Appointment

A scheduled booking for an eye test.

Prescription

A medical record containing vision correction details for a user.

Product

Items such as glasses, frames, and contact lenses available for sale.

Shopping Cart

A temporary list where users add products before purchasing.

Order

A confirmed purchase made by a user.

Order Tracking

The process of monitoring the status of an order.

Database

A system used to store user, product, prescription, and order data.

Authentication

The process of verifying user identity using login credentials.

Authorization

The process of controlling access to system features based on user roles.

User Interface (UI)

The visual part of the system that users interact with.

Frontend

The client-side of the system (what users see and interact with).

Backend

The server-side that handles logic, database operations, and processing.

Web Browser

Software used to access the system (e.g., Chrome, Edge).

Server

A system that hosts the web application and processes requests.

HTTP/HTTPS

Protocols used for communication between client and server (HTTPS is secure).